

Michelson Interferometry: Experiment I

Connor Fricke (fricke.59@osu.edu)

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Abstract—The famous Michelson Interferometer instrument as used in the Michelson-Morley experiment (See Ref. [1]) can be used for many optical measurement tasks, including the measurement of the wavelength of a laser, the separation between nearby spectral lines of a light source, and the index of refraction of a transparent material. Three sub-experiments are followed here, producing a HeNe laser wavelength measurement of $\lambda = 627 \pm 1$ nm, a Thallium lamp wavelength of $\lambda = 530 \pm 30$ nm, a Mercury yellow doublet spectral line separation of $\Delta\lambda = 2.07 \pm .03$ nm, and an index of refraction for a glass slide, $n = 1.38 \pm 0.05$. These measurements were each obtained within $\pm 6\sigma$ of the expected values, proving the effectiveness of the interferometer for optical measurements.

I. INTRODUCTION

The Michelson interferometer is an optical instrument often used to demonstrate the wave nature of electromagnetic light through the generation of interference patterns created by coherent interactions of two different wave sources. The primary function of the interferometer involves splitting a beam into two different paths of different lengths, allowing for the separate manipulation of these beams to perform various interferometry tasks (See Ref. [2]).

The interferometer was originally used in the famous Michelson-Morley experiment, which disproved the theory that light waves propagate through an invisible “ether” (See Ref. [1]). The experiment aimed to detect the relative motion of the Earth through the ether which would result in a change of the speed of light. The paper concluded that there was no detectable change in the speed of light from Earth’s relative motion, disproving the existence of the ether. It also paved the way for Einstein’s theory of special relativity.

The instrument is designed to provide extremely fine control over the path of each light beam. This control allows for intentional interference patterns to be observed from the recombination of the split beams. These interference patterns can be used to accurately predict beam wavelengths, wavelength separations of beam components, or the index of refraction of some transparent substance through which a beam passes.

There are three sub-experiments with the following objectives:

- 1) Measure wavelength, λ , of the Helium-Neon laser and of the Thallium green line.
- 2) Measure wavelength separation, $\Delta\lambda$, of the Mercury yellow doublet.
- 3) Measure the index of refraction, n , of a transparent material such as mica, quartz, or glass.

There are also a few secondary objectives of the experiment as a whole:

- 1) Demonstrate the wave nature of light with diffraction patterns.
- 2) Use Snell’s law to manipulate the path of a beam of light.

II. SAFETY CONSIDERATIONS

This experiment requires the use of lasers, meaning that proper care must be taken to avoid harm to participants. While the lasers that are part of this lab are not powerful enough to induce blindness, it is still necessary to take precautions to limit eye and skin contact with radiation emitted by the laser.

Ohio State provides students and instructors with a required course on Laser Safety through *BuckeyeLearn*. This one hour and forty minute course should be completed by all participants prior to usage of lasers in the lab (See Ref. [3]).

For this experiment, the laser beam should remain within the plane of the interferometer itself. To avoid looking into the laser, participants should avoid placing their head near the plane of the interferometer at all times, and instead observe the interferometer from above. The interferometer should be placed on a flat, stable surface. Rings, watches, and other jewelry should be removed to avoid specular reflections of the laser out of the plane of the instrument.

Standards of procedure should be followed carefully during the initial setup and alignment of the interferometer before the emitting device is fixed in place. Avoid aiming the laser upwards from the optics bench.

III. HELIUM-NEON AND THALLIUM

A. Theory

From a singular point source S , the Michelson interferometer generates two “virtual” point sources S'_1 and S'_2 via coordinated reflection and transmission of two legs of a split beam. Because of the varying path lengths of these beams, the virtual sources have a slightly different distance from the observer. The Michelson interferometer setup is shown in Figure 1. The separation between the virtual sources is given by

$$\delta' = 2\delta = 2(L_2 - L_1) \quad (1)$$

where L_1 is the distance between beam splitter and the mirror (M_1) which receives the light *reflected* by the beam splitter. L_2 is the distance between the beam splitter and the mirror (M_2) which receives the light *transmitted* by the beam splitter. Generally, there is fine motor control over one mirrors of the interferometer which allows for very slight adjustments to either L_1 or L_2 (in this case, L_2) [4].

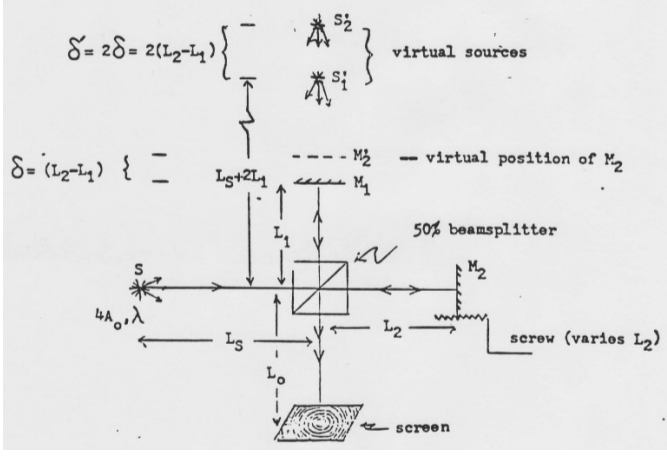


Fig. 1. Michelson Interferometer. L_S , L_O , L_1 , and L_2 are the optical paths indicated. (See Ref. [4]).

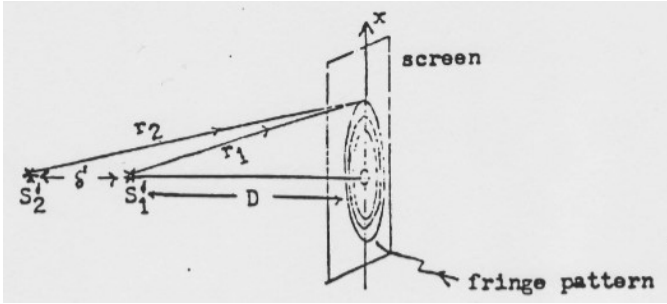


Fig. 2. Fringe pattern generated by virtual sources S_1' and S_2' . $D = L_O + L_S + 2L_1$; $\delta' = 2(L_2 - L_1)$. (See Ref. [4]).

Since the beam which follows L_2 must pass through the beam splitter three times, while the beam which follows L_1 must pass through only once, a second plate of identical thickness and material called a “compensator” is placed along the path of L_1 to eliminate the need to account for the thickness of the beam splitter [2].

The location of bright and dark fringes are given by the following equations as they relate to Figure 2.

$$|r_2 - r_1| = n\lambda \quad (2)$$

where $n = 0, 1, 2, \dots$ for bright fringes. For the dark fringes,

$$|r_2 - r_1| = n\lambda/2 \quad (3)$$

where $n = 1, 3, 5, \dots$. Since r_1 and r_2 can be calculated from known path lengths, accurate measurement of the location of the bright and dark fringes allows for an accurate prediction of the wavelength of light through these relationships (See Ref. [4]).

As the movable mirror is adjusted back and forth to change δ' , a number of rings will be seen to be either entering or exiting the center point of the “bullseye” fringe pattern. A more direct measurement of the wavelength of the source laser is given by

$$\lambda = \frac{2\Delta d}{N} \quad (4)$$

where λ is the laser wavelength, N is the number of fringes which pass by a fixed point on the screen, and Δd is the

distance moved by the adjustable mirror (See Ref. [5]).

B. Methods

The most time-consuming step of the experiment may be the alignment stage. The helium-neon laser should be placed into the laser mount and then turned on. Alignment of the mirrors must be done precisely and according to the standard of procedure so that the instrument can direct the laser beam through the beam splitter and mirrors properly. Instructions for the alignment stage can be found in Section 4 of Ref. [5]. It may be helpful to place a sheet of paper where the fringe pattern is expected to appear during the initial alignment before setting up the observing camera. Once the instrument is correctly aligned, a ring of bright and dark fringes will appear on the sheet of paper like the one shown in Figure 2.

Accurate observation and measurement of the interference patterns can be done with the ThorLabs DCC1545M USB camera or the ThorLabs DET36A photodetector (See Ref. [6] for operation manual). The video feed from the camera paired with the *ImageJ* software allows for the plotting of pixel intensity against pixel location, for example. The photodetector paired with an iris blocking all but the central fringe can accurately count the number of fringes observed as the mirror is adjusted.

To measure the wavelength of the HeNe laser, the process is as follows (See Ref. [5]):

- 1) Slowly adjust the micrometer which controls the position of the adjustable mirror
- 2) Choosing a fixed point (either a pixel on the video feed or on the paper where the fringes appear), count the number of fringes which pass by that point across the range of the micrometer. Note the micrometer adjustment value each time a fringe passes this point.
- 3) Depending on the direction that the mirror is moving, circular fringes will either enter or exit at the center of the pattern.
- 4) Repeat the procedure several times to minimize errors and to obtain an average and standard deviation.
- 5) Using equation (4), calculate the laser wavelength and propagate your error to obtain an uncertainty on λ .

C. Analysis & Interpretation

For measuring the wavelengths of both the HeNe laser and the thallium lamp, a table of number of fringes, N , and equivalent distance, Δd is produced. For the HeNe laser, we move the mirror such that a constant number of fringes passed and noted the distance. For the Thallium lamp, we instead moved the micrometer a fixed distance ($20 \pm 2 \mu\text{m}$) and record the number of fringes. In short, N is the independent variable for the HeNe laser, but Δd is the independent variable for the Thallium lamp.

According to equation (4), plotting Δd vs. N (HeNe) means that the slope of the resulting fit is $\lambda/2$. See Figure (3) for this plot. A weighted fit of *raw* data produces a fit parameter of

$$\lambda_{\text{measured}} = 658 \pm 1 \text{ nm.}$$

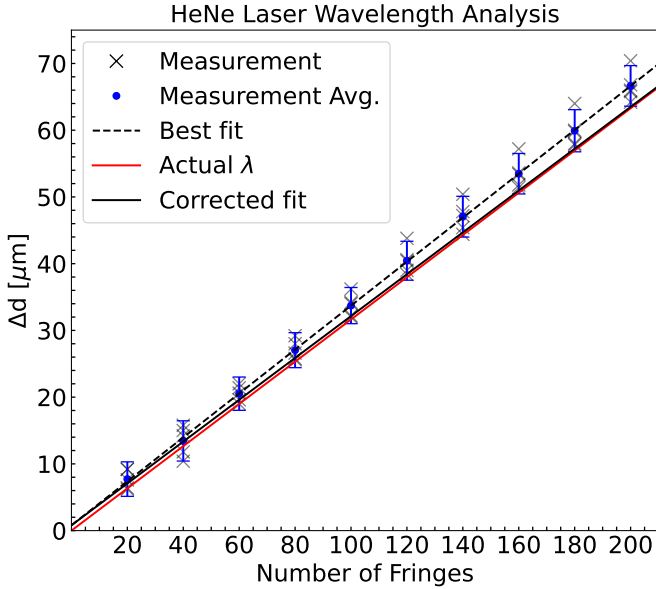


Fig. 3. Distance moved by the mirror plotted against number of fringes counted. Raw data is plotted as points, with the best fit ($\tilde{\chi}^2 = 0.257$) overestimating the true slope. Accounting for the directional bias and shifting data points by 1 fringe for every 20, the black line is created, which is a much more accurate prediction of the true HeNe wavelength, $\lambda = 632.8$ nm. For the corrected fit, ($\tilde{\chi}^2 = 0.99$), the prediction was $\lambda = 627 \pm 1$ nm.

Considering this is an overestimate of the expected value, $\langle \lambda \rangle = 632.8$ nm, it would be a reasonable statement that counting fringes is difficult, which introduces a *bias* into our raw data. To explain this *directional bias*, it can most simply be stated that it is far easier to miss fringes while counting rather than to over-count.

Let's assume, for the sake of testing this bias, that we missed one of every 20 fringes or so, such that we add in a shift of +1 for each $N = 20$. Thus our values of N become $N = \{21, 42, 63 \dots\}$ as opposed to $N = \{20, 40, 60 \dots\}$. Fitting this newly “corrected” data does indeed produce more reasonable results.

$$\tilde{\lambda}_{measured} = 627 \pm 1 \text{ nm} \quad (5)$$

Our measured value with the directional bias accounted for puts the expectation value within $\pm 6\sigma$. Note that for the uncorrected data, the error bars are quite large. However, because the fit is so close to this data, the error on the fit parameter is still quite small at only ± 1 nm, which was obtained from diagonalization of the covariance matrix.

data	$\tilde{\chi}^2$
original	0.257
corrected	0.99

The $\tilde{\chi}^2$ values suggest error bars are overestimated for the initial fit. The corrected $\tilde{\chi}^2$ value is obtained using the residuals of the corrected fit with the original data, thus it increases but remains near unity.

For the Thallium lamp, video recordings were used to slow down the motion of the fringes, though the data still displayed quite a large statistical variance in the number of fringes passed

over multiple trials due to low frame rate and vibrations of the optical bench, interferometer, or the micrometer. Additional sources of error include the systematic uncertainty in the width of the distance interval of the mirror adjustment ($\Delta d = 20 \pm 2 \mu\text{m}$). Counting of fringes for 15 different recordings produced the table below. One measurement (marked with *) was dropped because of the low quality and inconsistency of the recording.

range (μm)	Δd (μm)	N_1	N_2	N_3	N_4	N_5
0 $\rightarrow$ 20	20 ± 2	63	63	69	70	70
20 $\rightarrow$ 40	20 ± 2	68	72	69	60*	69
40 $\rightarrow$ 60	20 ± 2	72	60	65	73	71

All 14 remaining measurements in the table are technically recordings of the same point. Rather than performing a fit, a simple calculation (with error propagation of σ_N and $\sigma_{\Delta d}$) produces a measurement of

$$\lambda_{measured} = 580 \pm 30 \text{ nm} \quad (6)$$

which is an obvious overestimate of the expectation value of $\langle \lambda \rangle = 535$ nm.

By similar analysis to the previous measurements (assuming we successfully counted 10 of every 11 fringes that pass), we can illustrate the same *directional bias*. By multiplying each measurement by a factor of 1.1, the “corrected” data produces an estimate of

$$\tilde{\lambda}_{measured} = 530 \pm 30 \text{ nm} \quad (7)$$

Unfortunately, the error bars are still quite large due to the unavoidable systematic uncertainties created by existing near the interferometer during the experiment, which results in a large variance in the measurements in the table.

IV. MERCURY DOUBLET

A. Theory

The Mercury lamp actually produces two spectral lines with very similar wavelengths in the visible range of the spectrum. The interferometer can be used to accurately calculate the wavelength separation between these waves, often referred to as the yellow Hg *doublet*.

The main mechanism of this experiment relies on the fact that when two waves of very similar frequency are superimposed, there are periods in which the waves are *in phase*, and their amplitudes combine coherently in what is called *constructive interference*. Of course, since there is a small difference in wavelength, there are other regions in which the same waves are *out of phase* with each other, and their amplitudes cancel each other out in what is called *destructive interference*. This is the mechanism which creates interference fringes (bright and dark rings) which appear similar to the previous experiment.

Let's assume we have two wavelengths, λ_1 and λ_2 such that $\lambda_2 = \lambda_1 + \Delta\lambda$ and $\frac{\Delta\lambda}{\lambda_1} \ll 1$. Choosing an angle, θ , at which some dark fringe appears, and with a known difference between arm lengths d , one can write

$$2d \cos \theta = m_1 \lambda_1 = m_2 \lambda_2 \quad (8)$$

Adjusting the micrometer will bring the waves in and out of phase at θ . When the pattern returns to the high contrast state as before, the new arm length difference is d' , giving $\Delta d = d' - d$. The wavelength separation can then be calculated, given by

$$\Delta\lambda = \frac{\lambda_1 \lambda_2}{2\Delta d \cos \theta} \quad (9)$$

By looking at fringes near the center (small θ , $\cos \theta \approx 1$), equation (6) simplifies to

$$\Delta\lambda = \frac{\lambda_{avg}^2}{2\Delta d} \quad (10)$$

where λ_{avg} is the average wavelength of the doublet. (See Ref. [7]).

B. Methods

The observational instrument for this experiment should be the *ThorLabs* camera (not the photodetector) placed very close to the interferometer. The alignment process is very similar to the Helium-Neon setup, however this time the source will be a Mercury lamp instead of the previous laser. Since the lamp is less accurately approximated as a point source (and less harmful to the eye), it is helpful to look through the interferometer while adjusting the tilt controls of the mirrors so that the images of the lamp filament overlap as much as possible. Successful alignment will produce a similar bullseye pattern to before. It should appear multicolor, but mostly blue (though we are concerned with the *yellow* doublet, these are not the only spectral lines produced by the Mercury lamp).

A filter should be placed over the opening of the light source so that only the yellow doublet propagates through the interferometer. The bullseye will now be yellow.

As the position of the mirror is adjusted, the overall *contrast* of the interference pattern will cycle from high to low. A high-contrast pattern will look like very bright and dark rings, while a low-contrast pattern may look like no fringe pattern at all.

Recording data is a straightforward process of rotating the micrometer controlling the mirror position and noting the reading after each contrast cycle. Of course, it is nearly impossible to pick the precise point of lowest or highest contrast, thus we expect most measurement error of Δd will come from this characteristic of the process. The precise “location” of the low-contrast state is much more easily observed than the high-contrast state, since there is a moment when the fringe seems to disappear entirely at this point; therefore, cycles will be recorded at each low-contrast state.

C. Analysis & Interpretation

Data was collected over three repeated trials of five contrast cycles, recording the micrometer reading as close as possible to the lowest contrast state. The error in our ability to choose the precise point of lowest contrast should be reflected by the statistical uncertainty of repeated measurements. The following table includes these measurements. N is the number of cycles observed, and d_i is the reading on the micrometer (converted to μm) for trial i . The spacing of the smallest markings on the micrometer are equivalent to $2 \mu\text{m}$, thus

systematic uncertainty on each measurement of the table is $\pm 1 \mu\text{m}$.

N	$d_1 (\pm 1 \mu\text{m})$	$d_2 (\pm 1 \mu\text{m})$	$d_3 (\pm 1 \mu\text{m})$
1	56	58	60
2	140	138	140
3	218	220	220
4	300	300	300
5	380	380	380

To equate all rows of this data, we first subtract the values of the first row from each column to remove the non-zero intercept, accounting for the fact that the zero reading on the micrometer does not correspond to a low contrast state. Then, for each row where $N > 1$, dividing d_i by N produces measurements which are theoretically equivalent to each other. The mean and standard deviation of these values gives

$$\Delta d = 81 \pm 1 \mu\text{m}$$

Systematic uncertainty ($\sigma_{sys} = \pm 1 \mu\text{m}$) is added with statistical uncertainty ($\sigma_{stat} = \pm 1 \mu\text{m}$) in quadrature. Given known values of the spectral lines $\lambda_{<} = 576.9598 \text{ nm}$ and $\lambda_{>} = 579.0663 \text{ nm}$, we can calculate $\lambda_{avg} = 578.01305 \text{ nm}$. Using these values with equation (8) and propagating error yields

$$\Delta\lambda = 2.07 \pm 0.03 \text{ nm}. \quad (11)$$

which is within $\pm 1\sigma$ of the expected value of $\langle \Delta\lambda \rangle = 2.1 \text{ nm}$

V. INDEX OF REFRACTION

A. Theory

Index of refraction, n , is a material property which influences how much light is bent as it passes through the boundary of a substance. It is a ratio of the apparent speed of light in vacuum to the speed of light in the material.

$$n = \frac{c}{v} \quad (12)$$

A higher refractive index thus corresponds to slower light wave propagation in the medium. At the boundary between two mediums with different values of n , the path of the light is bent. For light passing from an n_1 medium to an n_2 medium, the angle of incidence, θ_1 , and angle of refraction, θ_2 are related by Snell's law:

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1} \quad (13)$$

θ_1 and θ_2 are given by the angle between the path of the light and the normal direction of the interface between mediums.

Clearly, the placement of a plate such as the one in Figure 3 would produce a predictable manipulation of the path length if the plate thickness and refractive index are known. If refractive index is an unknown, then measuring the resulting path difference with the interferometer allows for a calculation of the unknown, n . This is precisely the objective for this experiment.

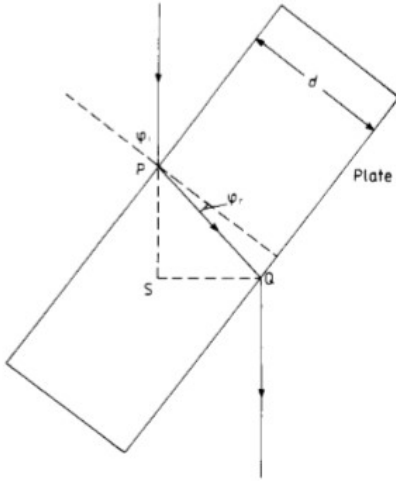


Fig. 4. Details of the light path through a plate. Knowing the thickness of the plate and the angle of incidence allows for a measurement of the refractive index. (See Ref. [8]).

The refractive index of a plate placed along the path of the beam traveling between the beam splitter and the fixed mirror is given by

$$n = \frac{(2t - N\lambda)(1 - \cos \theta)}{2t(1 - \cos \theta) - N\lambda} \quad (14)$$

where λ is the laser wavelength, t is the thickness of the plate of glass, mica, or quartz, N is the number of fringes passed by, and θ is the angle rotated, in radians (see Ref. [5]).

B. Methods

Similar to the first part of this experiment (the HeNe laser), this procedure requires counting the number of fringes which pass by a particular point on the screen as the Michelson interferometer is adjusted. This time, instead of varying path length by moving the adjustable mirror with a micrometer, the rotational position of the plate in question is adjusted. This changing θ produces measurable effects on the final path length of the beam which travels through the plate, producing the expected interference patterns.

The experimental procedure is outlined as follows:

- 1) Measure the thickness of the plate of mica, quartz, or glass that is being tested.
- 2) After alignment of the interferometer so that the bullseye pattern appears on the screen, the plate should be placed in the mount.
- 3) Perform an initial alignment of the plate with the rotational adjustment handle so that it starts normal to the beam path.
- 4) Begin finely adjusting the angle of the plate. Make a note of the angle of incidence, θ , each time that a bright fringe passes the chosen fixed point on the screen, keeping track of the total number of fringes that have passed, N .
- 5) Repeat the procedure several times to obtain a statistical uncertainty to measurements.

- 6) Using the measured values of t , N , θ , and the known value of λ , calculate an estimate of n and propagate the existing errors on t , N , and θ .

C. Analysis & Interpretation

Analysis of results for this sub-experiment proved to be quite difficult. Because of the inaccuracies of counting by eye, a photo-detector and universal counter setup was used to record the number of fringes which passed over a particular angular interval.

The protractor on the slide mount was divided into 50, with each division having 10 subdivisions, thus a division is equivalent to $\frac{2\pi}{50}$ radians, a subdivision $\frac{2\pi}{500}$ radians, and the error on each angular measurement is $\pm \frac{2\pi}{1000}$ radians. The laser used for the experiment was the HeNe laser, with $\lambda = 632.8$ nm. A caliper was used to measure the thickness of the glass, obtaining

$$t = 1.02 \pm 0.03 \text{ mm.}$$

Number of fringes passed is reported for every 5 subdivisions of the protractor in the following table:

division ($2\pi/50$ rad)	θ (± 0.006 rad)	N_1	N_2	N_3
-1.5	0.754	0	0	0
-1.0	0.691	58	56	60
-0.5	0.628	119	110	109
0.0	0.565	155	159	157
0.5	0.502	198	208	192

Error propagation must be performed carefully in this case, such that the errors in N , t , and θ are accounted for. The error on refractive index for a particular set of measurements with errors σ_N , σ_t , and σ_θ is given by

$$\sigma_n^2 = \left(\frac{\partial n}{\partial N} \right)^2 \sigma_N^2 + \left(\frac{\partial n}{\partial t} \right)^2 \sigma_t^2 + \left(\frac{\partial n}{\partial \theta} \right)^2 \sigma_\theta^2 \quad (15)$$

Each row in the table produces an individual result of n with a given σ_n . To obtain a final prediction, we average the results and add their errors in quadrature, obtaining a final prediction

$$n = 1.38 \pm 0.05 \quad (16)$$

such that our result is within 3σ of the expectation value for glass, $n = 1.51$ (See Ref. [9]).

VI. CONCLUSIONS

The Michelson interferometer has successfully been employed in the measurement of four optical quantities via exploitation of the wave nature of light. The wavelength of both a HeNe laser and a Thallium lamp were measured at

$$\begin{aligned} \lambda_{\text{HeNe}} &= 627 \pm 1 \text{ nm } (\pm 6\sigma) \\ \lambda_{\text{Thallium}} &= 534 \pm 30 \text{ nm } (\pm 1\sigma) \\ \Delta\lambda_{\text{Hg}} &= 2.07 \pm 0.03 \text{ nm } (\pm 1\sigma) \\ n_{\text{glass}} &= 1.38 \pm 0.05 (\pm 3\sigma) \end{aligned}$$

As noted above, each measurement was taken within $\pm 6\sigma$ of the expectation values,

$$\begin{aligned}\langle \lambda_{HeNe} \rangle &= 632.8 \text{ nm} \\ \langle \lambda_{Thallium} \rangle &= 535 \text{ nm} \\ \langle \Delta \lambda_{Hg} \rangle &= 2.10 \text{ nm} \\ \langle n_{glass} \rangle &= 1.51\end{aligned}$$

Improvements could be made to the experiment to obtain more accurate results, such as electronically regulated fine motor control over the mirror position as well as the glass slide angle, which would limit statistical uncertainty introduced by hand-control of the instrumentation. The photodetector could be employed in the counting of the HeNe laser fringes to obtain results quicker and more precisely. In general, though, the interferometer has proven it's capabilities for accurate estimates of wavelengths and indices of refraction.

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